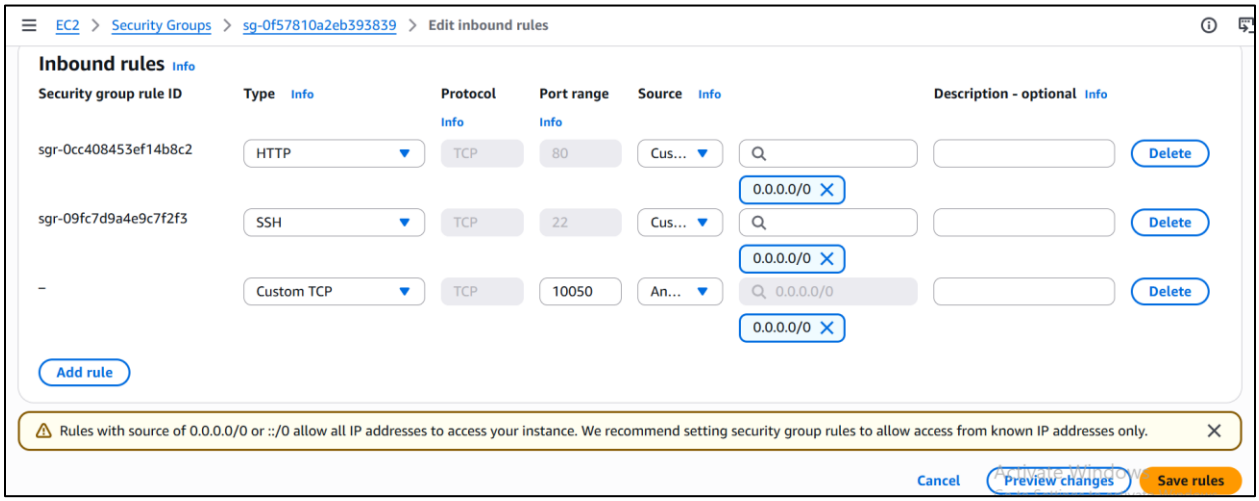
AWS EC2 Multi-Service Deployment Lab (NGINX, MongoDB, Zabbix)

AWS EC2

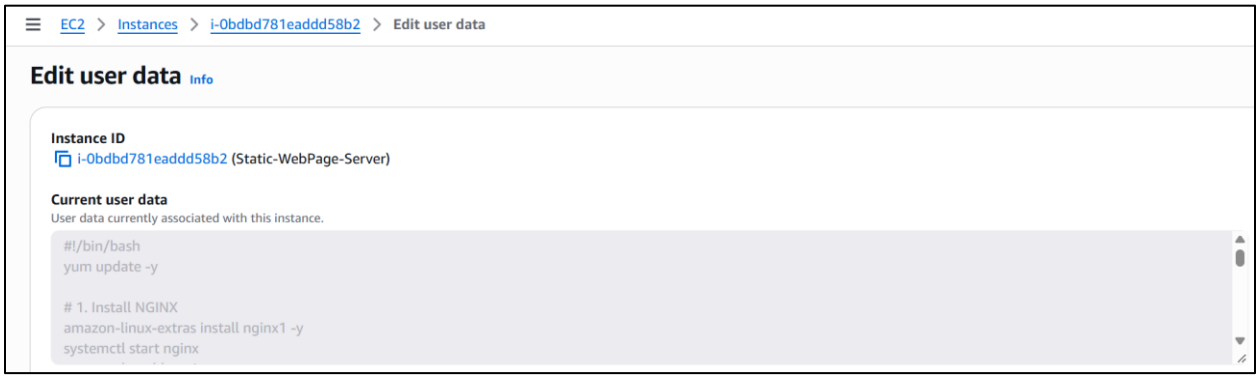
Lab: User Data Script

- Prepare EC2 User Data script for **Amazon Linux 2** using **AI** so that:
  - Installs **NGINX**
  - Installs **MongoDB**
  - Installs **Zabbix Agent** (lightweight monitoring tool)
  - Automatically starts all services
  - Outputs a confirmation log.
- Ensure the security group attached to the EC2 has these **open ports**:
  - **TCP 22** (SSH)
  - **TCP 88** (NGINX)
  - **TCP 10050** (Zabbix Agent default)
- Connect to the instance via **SSH** and check that the above services are running.

EC2 (Security Group)



EC2 (Instances)



NGINX ACTIVE

```
Complete!
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.ser
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-10-03 07:22:55 UTC; 30s ago
   Main PID: 3578 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 59ms
     CGroup: /system.slice/nginx.service
             ├─3578 "nginx: master process /usr/sbin/nginx"
             ├─3579 "nginx: worker process"
             └─3580 "nginx: worker process"
```

MONGOD ACTIVE

```
Installed:
  mongodb-database-tools-100.19.1-1.x86_64                mongodb-mongosh-2.12.0-1.el8.x86_64
  mongodb-org-7.0.43-1.amzn2023.x86_64                    mongodb-org-database-7.0.43-1.amzn2023.x86_64
  mongodb-org-database-tools-extra-7.0.43-1.amzn2023.x86_64  mongodb-org-mongos-7.0.43-1.amzn2023.x86_64
  mongodb-org-server-7.0.43-1.amzn2023.x86_64             mongodb-org-tools-7.0.43-1.amzn2023.x86_64

Complete!
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl start mongod
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl enable mongod
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
     Loaded: loaded (/usr/lib/systemd/system/mongod.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-10-03 07:33:39 UTC; 25s ago
       Docs: https://docs.mongodb.org/manual
   Main PID: 4389 (mongod)
     Memory: 150.1M
        CPU: 798ms
     CGroup: /system.slice/mongod.service
             └─4389 /usr/bin/mongod -f /etc/mongod.conf

Oct 03 07:33:39 ip-172-31-24-195.ec2.internal systemd[1]: Started mongod.service - MongoDB Database Server.
Oct 03 07:33:39 ip-172-31-24-195.ec2.internal mongod[4389]: {"t":{"$date":"2026-10-03T07:33:39.925Z"},"s":"I",  "c":"CONTROL",  "id":74
lines 1-12/12 (END)
```

ZABBIX ACTIVE

```
Nothing to do.
Complete!
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl start zabbix-agent
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl enable zabbix-agent
[ec2-user@ip-172-31-24-195 ~]$ sudo systemctl status zabbix-agent
● zabbix-agent.service - Zabbix Agent
     Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-10-03 07:18:06 UTC; 19min ago
   Main PID: 2360 (zabbix_agentd)
      Tasks: 6 (limit: 1059)
     Memory: 2.6M
        CPU: 235ms
     CGroup: /system.slice/zabbix-agent.service
             ├─2360 /usr/sbin/zabbix_agentd -c /etc/zabbix/zabbix_agentd.conf
             ├─2361 "/usr/sbin/zabbix_agentd: collector [idle 1 sec]"
             ├─2362 "/usr/sbin/zabbix_agentd: listener #1 [waiting for connection]"
             ├─2363 "/usr/sbin/zabbix_agentd: listener #2 [waiting for connection]"
             ├─2364 "/usr/sbin/zabbix_agentd: listener #3 [waiting for connection]"
             └─2365 "/usr/sbin/zabbix_agentd: active checks #1 [idle 1 sec]"

Oct 03 07:18:05 ip-172-31-24-195.ec2.internal systemd[1]: Starting zabbix-agent.service - Zabbix Agent...
Oct 03 07:18:06 ip-172-31-24-195.ec2.internal systemd[1]: Started zabbix-agent.service - Zabbix Agent.
[ec2-user@ip-172-31-24-195 ~]$
```

Testing



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.